ETH zürich | Physikalisches Praktikum der ETH Zürich | Physics Lab

Name 1	Datum:
Name 2	Platz-Nr:

09 - absolute Zero

Calibration of the pressure sensor

Calibration point at ambient pressure

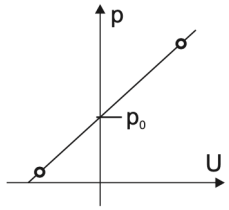
(uncorrected barometer reading)	(temperature at barometer)	(corrected barometer reading)
p_L= mm Hg (t_L)	air temp t_L= °C	p_L= Torr
p_L= Pa	U_L= mV	

Calibration point at low pressure

p_t= mbar	p_t= Pa	U_t= mV

Sensor characteristics: $p = p_0 + CU$:

p_0=	C=



Determination of the absolute zero

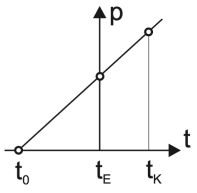
At the temperature of boiling water

t_K= °C	U_K= mV	p_K= mbar

At the temperature of ice water

t_E= °C	U_E= mV	p_E= mbar

Approximation for the temperature of the absolute zero:



t'$_0$ =.....

Volume of the empty space / volume of the glass bulb: $V_s/V = \varepsilon = 1 \cdot 10^{-3}$
Cubic expansion coefficient of glass: $\gamma = 1.0 \cdot 10^{-5}\,°C^{-1}$

a=	b=	c=

Absolute zero of temperature:

t_0 =.....

Determination of the temperature of liquid nitrogen

U_{LN2}= mV	p_{LN2}= Pa	t'$_{LN2}$ = °C
corr. coeff A=	t_{LN2} = °C	